

Naturalising deconstruction

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Abstract. Most contemporary readings of Derrida's work situate it within a transcendental tradition of philosophical enquiry explicitly critical of naturalistic accounts of knowledge and mind. I argue that Derrida provides the naturalist with some of the philosophical resources needed to rebut transcendental critiques of naturalism, in particular the phenomenological critiques which derive from Husserl's philosophy. I do this by showing: a) that Derrida's account of temporality as *differance* undermines phenomenological accounts of the meaning of naturalistic theories and assumptions; and b) that it is itself both usable and interpretable within the naturalistic framework of current cognitive science.

Contemporary naturalists hold that the current best scientific accounts of the world should inform or constrain philosophical accounts of the world. As a corollary, naturalists are anti-foundationalists: rejecting claims to *a priori* truth or insight into transcendental conditions of possibility that are immune to empirically motivated conceptual change.

Naturalism is contested by contemporary forms of subjectivist phenomenology and transcendental philosophy which hold that there are invariants of objectivity or intersubjective understanding presupposed in any scientific theory or discourse. Derridean deconstruction is often treated as anti-naturalistic by default of its relation with the transcendental and phenomenological tradition. Despite Derrida's early critique of Husserl's phenomenology, leading commentators like Rodolphe Gasché imply that deconstruction radicalises the phenomenological critique of naturalism.¹ On this widely held view, Derrida is firmly in the transcendental camp; a philosopher for whom thought and experience are constituted by 'non-causal conditions of possibility' recalcitrant to conceptual thought and, by extension, any form of ontological account derived from science.²

However, this 'quasi-transcendentalist' approach to deconstruction fails to recognise that Derrida's work supports the assimilation of phenomenologically-derived conceptions of meaning within a naturalistic framework by exhibiting points at which Husserl's phenomenological descriptions undermine his methodological exceptionalism. What distinguishes transcendental phenomenology from other disciplines is that it proposes to begin with the intuitive data of pure experience rather than with presumptive

claims about entities which transcend such experience. Derrida's position in Chapter Five of *Speech and Phenomena*, 'Signs and the Blink of an Eye' (*Le sign et le clin d'oeil*),³ in particular, implies that the constitutive conditions of temporal experience must be non-intuitable or recalcitrant to phenomenological method.

In standard readings of Derrida this conclusion is taken to entail some deeper set of quasi-transcendental conditions – indicated by neologisms such as 'differance', 'iterability', or 'trace' – which, though 'non-phenomenologisable' (in Gasché's terminology), nonetheless mark out conditions of possibility for thought or scientific conceptuality as such. The standard reading thus preserves transcendental schemas of thought while dispensing with subjectivist modes of transcendental reflection. I argue, on the contrary, that deconstruction is able to mobilise concepts and arguments which have a provenance in transcendental philosophy, but in a way that opens them up to naturalistic construal.⁴ Notions like differance may, then, be of value in theorising the way in which non-phenomenologisable processes which give rise to experience and meaning impact upon representational or intentional contents.⁵

In what follows I will set out a somewhat idealised geography of contemporary naturalism in the philosophy of mind – embracing intentional realism, eliminativism, dynamicist approaches, and interpretationism – and explore their points of conflict with transcendental phenomenology (part 1). In part 2, I will show how deconstruction can be used to rebut the phenomenological critique of naturalism. Part 3 describes how naturalised deconstruction can inform an interpretationist approach to intentional content.

1. Naturalism and phenomenology

Naturalism is most commonly presented as an approach to issues of mind, knowledge and cognition. Naturalists of different philosophical hues believe that mental phenomena such as our capacity to represent the world or possess phenomenal consciousness should be understood within the ontological framework of contemporary natural science. The naturalist wants to understand how phenomena like cognition and emotion 'fit' into a causal-physical account of the world.

Many of the differences between leading naturalists such as Daniel Dennett, Paul and Patricia Churchland, or Jerry Fodor arise from disputes about the proper form and epistemic status of naturalistic theories of content or intentionality.

Fodor is an intentional realist who believes that a naturalistic theory of content should specify causal or functional sufficient conditions for a physical

state having a particular psychological role and a particular propositional content: e.g. being *the belief that Lima is the capital of Peru*. He proposes that the compositional structure implied by natural language expressions like 'Lima is the capital of Peru' is reflected in a 'language of thought' (LOT) which explains the productivity and systematicity of cognition in terms of syntactical properties of identically repeatable LOT 'expressions'. LOT exhibits what Francisco Garzón refers to as *classical constituency*: complex mental representations being compounded from syntactically simple tokens whose content is repeated identically, without regard to sentential or functional context.⁶ The content of these constituents, according to Fodor, is to be accounted for in terms of causal dependencies between the representational tokenings and the environmental features which bring them about. Other representationalists look to the functional roles of representations or evolutionary considerations to distinguish between those causal relationships that are meaning-constitutive and those that are not.⁷

Eliminativists like the Churchlands hold that a truly explanatory account of the mind's representational capacities will not refer to intentional states (beliefs, desires, etc.), syntactically structured representations or propositional contents, but will be committed to completely different kinds of entities such as higher dimensional states of neural networks.⁸ Their approach can be subsumed into a dynamicist approach to cognition which explains behaviour in terms of 'global' properties of systems where a number of variables develop interdependently as a function of time.

Dennett has no problems with the kinds of explanatory mechanism postulated by eliminativists but argues that folk psychological usage of intentional concepts is ontologically neutral regarding the dynamic or computational nature of the mind/brain. His 'interpretationism' holds that intentional concepts like *belief* figure in explanations and predictions of behaviour which assume the rationality and cognisance of the agent under interpretation. When we view an agent in this way, we assume what Dennett refers to as the 'intentional stance'.⁹ For Dennett (as for Donald Davidson) a behavioural episode is indicative of intentionality only where – applying the principle of 'charity' – it can also be construed as appropriate or rational relative to the agent's environment. But intentional states obey such norms only when considered against the background of other intentional states.¹⁰ The moral, for both philosophers, is that no states of the world mean 'Lo, a rabbit' or (allowing for indeterminacy of interpretation) 'Lo, furry, large-eared quadruped' independently of their holistic interconnections with other intentional states and actions.¹¹

As Kathleen Akins observes, Dennett's interpretationism occupies a middle ground between realism and eliminativism: 'a commitment to the taxonomy of folk psychology but a denial that folk-psychological states are the correct

subject of a naturalistic theory'.¹² Interpretationism yields a naturalistic account of mind by an indirect route. It does not seek a smooth reduction of our intentional vocabulary to some physicalistically respectable idiom along realist lines, or seek to replace it altogether, as the eliminativists propose. Rather, it gives an account of the conditions under which an idealised interpreter would be warranted in construing a system as intentional. However, since these conditions involve intentionally characterised behaviour, any physicalistically respectable account which can explain how this behaviour is produced – by giving mechanistic accounts of the cognitive capacities or functions necessary to produce it, say – contributes to an explanation of the systematic dependence or 'supervenience'¹³ of the mental on the physical.

In the case of reductionist and eliminativist accounts of mind the incompatibility with transcendental phenomenology and related doctrines is stark. Transcendental phenomenology presupposes that any description of the world is a perspective of a transcendental subject or 'ego' which constitutes the sense of the world as an object of intentional experience.¹⁴ The constitution of sense or intentional content presupposes invariants, such as the synthesis of temporally contiguous phases of conscious experience and the lived experience of bodily situatedness and motility, which organise the 'appearing' of the objects towards which our experiences are directed.

For Husserl, the *meaning* of claims about physical entities of the kind that would play a role in any eliminativist or reductionist ontology can be adequately explicated only in terms of possible modes of awareness of them.¹⁵ These, it is claimed, can be described by undertaking a phenomenological *epoché* or reduction whereby ordinary assumptions about the objective being of objects of awareness are 'put out of play' and become the subject of phenomenological analysis. The 'world' of modern natural science, thus understood, is just one way of interpreting the culturally moulded 'life world' of perceptual objects, qualities and values.¹⁶ The 'naively objectivist' framework of naturalism is contingent and ultimately incomplete. The transcendental framework, by contrast, is presupposed in *any* stance towards a possible object of consciousness.

But a reductionist/eliminativist account of mind is committed to redescribing the totality of mental life without unreduced intentional notions. The notion of a belief being 'about' Lima (having that city as an intentional object) would simply not figure in a mature eliminativist account. Instead the eliminativist hopes to explain how purely structural or syntactic relations between organisms' neural states co-vary with salient features in the world (temperature, audible spectrum, etc.) in such a way as to prompt a (misleading) view of mental states as replete with semantic properties.¹⁷ The 'intentionality of experience' – so central to the phenomenological picture – would be treated

as a term in a naïve and ultimately unproductive folk theory rather than a privileged starting point in philosophical enquiry.

For proponents of phenomenological anti-naturalism like Matthew Radcliffe, this entails that any reductionist/eliminativist programme will be constitutively incomplete. If the objectivity of the world *consists* in it being an object for intentional consciousness (transcendence) the elimination of intentional discourse leaves the eliminativist incapable of describing the conditions under which her theory counts as objective or ‘scientific’. The reduction of intentional idioms to an objective naturalistic one (Fodor’s talk of causal dependences between LOT tokenings and environmental features, say) fares no better. For the phenomenologist the intentional relatedness of subject-to-object is conceptually prior to the objective and cannot be objectified without loss of intelligibility.¹⁸

Interpretationism is less obviously exposed to this criticism since it does not counsel replacement of intentional taxonomies with non-intentional ones. For the interpretationist, that an agent may be characterised as rationally engaged with their environment is sufficient and necessary for them having contentful states like beliefs. So in attributing these states one presupposes a comprehensive knowledge of the agent’s world. Interpretationism is thus a form of ‘externalism’. It holds that the contents of an organism’s intentional states are determined in large part by the things that it interacts with in its world rather than by ‘narrow’ psychological or phenomenological contents (which could be shared with physical or functional replicas in different environments).

This would render interpretationism incompatible with transcendental phenomenology were the latter committed to an internalist or individualist account of content. This is the most intuitive reading of Husserlian accounts of content since they require content to be accessible from a first-person point of view and to be independent of real physical contexts.¹⁹ However, the phenomenologist can also hold that the two theories are compatible but employ radically distinct notions, reserving the primary (‘world-constituting’) intentionality for the subject of phenomenology while claiming that interpretationism applies a theoretical construct which abstracts from its world-constituting role.

If, as Husserl claims, it is possible to suspend our commitment to the existence of transcendent objects like rocks or quarks whilst retaining the content of *rock* and *quark-directed* beliefs or perceptions, it can be objected that nothing precludes phenomenological investigation into activities like intentional interpretation: e.g. by explicating the meaning of the ‘common world’ shared by interpreter and interpretee. Interpretationism would then be in the same position of epistemic subordination as eliminativism and reductionism. Each

would address a philosophically ‘reconstructed’ notion of mental life whilst presupposing an unnaturalisable world-constituting intentionality.

The claim that naturalistic claims can be bracketed in phenomenological reflection presupposes that their sense or content persists after we have suspended any commitment to the truth claims of science or common sense. Just how this residuum is to be conceived is a live issue, both in contemporary phenomenology and Husserlian exegesis (e.g. is it to be conceived as a ‘static’ field of intuitable objects, meanings and essences or as cultural and habitual horizons of expectation which constitute the meanings of intentional objects over time?). It is not possible to adjudicate these issues here.²⁰ However, whether we privilege a static view of intentional content or a ‘genetic’, embodied view, or some synthesis of the two, the existence of a transcendental dimension of intentional experience requires that all intentional experiences a) have invariant or ‘essential’ features; b) some of these features are constitutive of the content of such experiences; and c) that constitutive features are possessed whether or not the claims subjected to *epoché* are true.

Any putatively constitutive feature whose nature depended on contingent and defeasible claims about transcendent entities could not elucidate the structural transcendence of the world considered as a necessary dimension of intentional consciousness. For example, according to Husserl it is possible that physical things might not have exhibited the mathematical relations expressed in modern physical theories *while still functioning as correlates of transcendental experience*.²¹ Thus Newtonian mass cannot be among the transcendentially constitutive conditions of experience. Otherwise, developments in physics since Newton’s time would have altered our view of the transcendental conditions of experience. The transcendent would have turned out to condition the transcendental (in violation of c) rendering a phenomenological account of essential transcendence strictly impossible.

Thus if a significant class of transcendental claims are predicated on properties or structures whose nature is (like mass) only determinable through the construction and testing of empirical theories, phenomenological ontology must be continuous with naturalistic metaphysics and lack methodological priority to it. Putatively transcendental conditions will have been shown to be admissible only as theoretical posits *within* the framework of empirical knowledge. In this case, no phenomenological position could claim epistemological independence or priority with regard to naturalistic theories of mind.

In the next section I shall use Derrida’s reading of Husserl to show that phenomenological temporality has the status of an empirical determinable and thus that the phenomenological account of transcendence derived from it cannot have epistemic priority to naturalistic claims regarding the mind.

2. Deconstruction *contra* phenomenology

Derrida's later views of meaning, time and the event draw substantially on Husserl's claim that we should understand the temporality of intentional experience as a differential relationship between memorisation and anticipation, presence and absence, rather than as an ordinal series of 'nows' or instants.²² Husserl sees this 'differing' as involving the immediate 'retention' of the past contents of perception – 'a sort of direct contact with the past in its own domain' – as well as a 'protention' or anticipation of what is to come.²³ The now-impression continuously passes into retention, accounting for the putative fact that we experience any temporal part of a perceived object in continuity with its just perceived temporal parts. Husserl thus combines phenomenological description with an explanation of why experience does not fracture into temporally isolated instants or mere associative complexes.²⁴

It is important that Husserl contrasts retention with ordinary reproductive memory. The content of retention is indubitably given whereas memories may misrepresent both the nature and the ordering of previous events (retention being non-representational).²⁵ The content of recollections, thoughts and beliefs regarding temporal orderings is parasitic on the duration constituted by retention-impression-protention. This has a transcendental role insofar as it renders intelligible the experience and thought of temporal relationships.²⁶

In 'Signs and the Blink of an Eye' Derrida argues that Husserl's continuist account of temporality undermines the primacy of phenomenological intuition by destabilising the distinction between the intuitive presence of a thing and a signifying or representational awareness of it. Since the temporal source point depends on a phase of experience which is necessarily displaced from that source, phenomenological evidence depends on two variants of the non-presence or 'repetition' of the present. The distinction between retention and reproduction has a 'common root' in the structure of the 'trace', which, according to Derrida, is 'more "primordial" [*originnaire*] than what is phenomenologically primordial'.²⁷ This qualifies the content of any experience as the structural possibility of a 'return' which modifies or 'supplements' its content or character. It is the fact of this non-identity or fold (*pli*) in time which allows each present to be 'retended' and grasped as an instance of the 'now' in general.²⁸

The trace is related to difference insofar as its content is never tokened in a single instance but is perpetually deferred through subsequent interpretations or re-presentations: '[Difference] is ... inconceivable as the mere homogeneous complication of a diagram or line of time, as a complex "succession"'. The supplementary difference vicariously stands in for presence due to its primordial self-deficiency'.²⁹ 'Trace' can also be regarded as another name

for what the later reading of Austin refers to as ‘iterability’. For a sign to function in normal or ‘serious’ contexts, Derrida argues, it must be repeatable (iterable) in contexts which alter its semantic value.³⁰ Any sign or contentful state must be recognisable independently of context in order to function in communication or representation, thus can be thought of as an ideal object or ‘type’.³¹ But its iteration cannot be the reinstantiation of an abstract object; for any significant particular – a word, a mental representation or a moment of experience – can always be detached from its context and ‘grafted’ into a new one in which it can mean differently.³²

As I will argue below, trace, differance and allied Derridean concepts like iterability provide a potentially fruitful vocabulary for addressing the systematicity and contingency of meaning. However, to make effective use of them we need to be clear about their theoretical status. Despite repeated hedges and qualifications, Derrida claims that they are non-subjective conditions of possibility for phenomenological consciousness and evidence. If this is correct, it must be possible to show (via some kind of transcendental argument) that Husserlian temporality presupposes the trace structure as its condition of possibility. However, though Derrida claims that the trace is prior to phenomenological conditions of possibility, he presents no argument in ‘Signs and the Blink of an Eye’ showing that the semantic openness indicated by the term is recalcitrant to phenomenological interpretation. As J. Claude Evans observes, Husserl’s account already entails the semantic instability of temporal consciousness and the interdependence of the phases of temporal consciousness.³³ This would be a problem were phenomenology committed to a punctual or ‘simple’ present, but this is precisely what Husserl’s continuist model of temporal experience is designed to replace.

It is possible, nonetheless, to buttress Derrida’s argument so as to establish that the constitutive structure of temporality resists phenomenological reflection. Here, the process of abstraction Derrida applies to Husserl claims in ‘Signs’ and elsewhere is the heuristic key to understanding how Derrida can point us beyond phenomenology. Texts such as “‘Genesis and Structure” and Phenomenology’ allude to a tension between methodology and ontology, implying that phenomenology cannot fix the meaning of cardinal terms such as ‘noema’ where such entities must, *by dint of their theoretical role*, elude subjective reflection.³⁴ It is important to recognise that such indeterminacy claims need not be established on transcendental grounds. They merely require an argument to the effect that some interpretation of a theory is incompatible with a particular doctrine of evidence for that interpretation. Where a phenomenological interpretation hinges on phenomenological methodology, this will follow if objects or relations posited in the former are demonstrably non-intuitable.

In the present case we begin with some modest constraints on the concept of intuition. Even if we concede grounds for disagreement about what it involves, certain entities must be conceded as non-intuitable within a phenomenological framework. For example, whereas each experience of a physical object can (for Husserl) be experienced adequately both as a real mental (*noetic*) event and as having a correlative ideal content or *noema*, the object – transcending any finite series of perceptions – is not given fully in this way. Its transcendence *just is* the possibility of being given under further perceptual or cognitive aspects, rather than existing as a determined thing-in-itself.³⁵ Similarly, certain mathematical and physical entities are, as Husserl plausibly claims, comprehensible only in terms of the theoretical structures in which they are posited and completely resist term by term reduction to phenomenological givens.³⁶ This is not a problem where we are dealing with entities with no constitutive role. However, as I have argued in the previous section, it would be a decisive problem if phenomenology were required to posit constitutive structures exhibiting a similar kind of transcendence.

The constitutive basis of temporal experience as described by Husserl is precisely a structure of this kind. Like the object of perception it can never be given fully in intuition. The analytical boundary between the impression and its retention is an abstraction. Since no impression exists in isolation from retentive or protentive phases of experience, the underlying structure of temporality must be treated as a continuum of relations in continuous alteration.³⁷ Every segment of the continuum is divisible, and at every grain we encounter phases ‘differing’ or modifying one another. It follows that ‘intuiting’ the structure of the temporal flux is as much an endless task as running through the perceptions of a transcendent physical object. While we may be non-inferentially aware of the passage of time in the empirical way we are aware of enduring objects or processes, no such possibility applies to the fine structure of lived time.³⁸

It could be objected that my argument trades on a simplified conception of phenomenological reflection. The fact that we cannot intuit a physical thing completely does not entail that we do not intuit it as something which transcends a finite train of experiences. Secondly, Husserl claims that there are forms of categorical or ‘essential’ intuition (*Wesensschau*). It could thus be objected that what is intuited is not the continuum *in itself* but the generative principle whereby any part of the continuum carries its retentive train.

However, if the structure of the temporal continuum is transcendent, there is no intuitive basis for assigning it a particular structure. The temporal flux remains as inaccessible to intuition as the fine structures of space or matter. Likewise the *Wesensschau* objection cannot resolve the problem for the phenomenologist. For if we allow that some principle of construction can be

apprehended in the coarse phenomenology of time it remains open whether and to what extent this captures the fine structure of temporal awareness.

The substantive grounds for preferring Husserl's continuist conception to, say, an atomistic or associationist one involve the epistemic role of the intuition of temporal order in securing the intelligibility of our ordinary temporal distinctions. If the flux was not continuous it would have discrete parts and discontinuities. It would then be hard to distinguish Husserl's account of our grasp of temporal relations from the associationist conceptions it is intended to replace. As David Wood shows, this is a decision which goes beyond the phenomenological data since the continuum postulated by Husserl can only be inferred from the need for some direct 'acquaintance' with temporal difference.³⁹

The retention-impression-protention trichotomy is a theoretical reconstruction of a process whose nature can no more be determined by phenomenological investigation than the phenomenology of motion can settle the issue between Newtonian or Einsteinian conceptions of mass. The *epistemological* status of Husserl's temporal flux is thus no different from that of other explananda in the natural sciences. The phenomenology of time is a good place to look if we wish to construct theories of temporal awareness. However, in lieu of phenomenological access to the processes underlying temporal awareness, other routes need to be considered. As Timothy van Gelder remarks in his essay 'Wooden Iron? Husserlian Phenomenology Meets Cognitive Science' there is an intriguing parallel here between the phenomenology of time and classical genetics. The latter set out general features of heredity and left molecular biology to specify 'the chemical compounds and processes which in fact have the general properties'.⁴⁰ Van Gelder illustrates how specification might proceed in the former case by using the Lexin model of auditory perception to reinterpret the Husserlian trichotomy in dynamical terms. Here, the path through the state space of a neural network designed to recognise patterns of variation in sound frequency is governed by current input and time-delayed connections between clusters of 'neurones'. This allows neurones in each cluster to acquire graded preferences for features of patterns (e.g. 'rising' as opposed to 'falling') during training. Thus a given sound will elicit chained responses which vary according to the pattern of frequencies preceding it. The system's trajectory through state space at any given moment thus bears the imprint of past 'experience' whilst remaining open to modification by subsequent 'experience'.

Van Gelder argues that this double dependency satisfies Husserl's requirement that retention and protention be 'built into' the present rather than merely associated with it. 'Retention' and 'protention' jointly reflect the way in which the system evolves in response to different sequences of auditory patterns.

However, there is an important revisionary aspect to his explanation. If the terms ‘retention’ and ‘protention’ refer at all, here, they pick out the system’s propensities or dispositions rather than quasi-perceptual intuitions of temporal difference. Indeed, for Van Gelder, retention and protention are properties of the *same* underlying disposition.

Van Gelder’s proposal demonstrates that Husserl’s temporal flux can be treated on a par with any object of empirical inquiry (though it is not necessary for my argument that his specific proposal be correct). Derridean deconstruction shows why there is nothing surprising or methodologically problematic in such an approach, since it warrants treating the phenomenology of time as a theoretically constructive enterprise with no serious differences from similar projects in the cognitive or natural sciences.

As the schematic argument pursued in the previous section implies, this levelling undermines claims for the methodological priority of transcendental phenomenology. For if temporality resists phenomenological explication, any notion that invokes it likewise resists phenomenological explication.

The phenomenological analysis of the transcendence of physical things is one of these since (like all phenomenological concepts) it depends upon the theory of temporality. Husserl understands the independence or objectivity of physical things in terms of a *synthesis* of experiences and anticipations which never give us the thing in its totality: “‘The’ thing itself is always in motion, always, and for everyone, a unity for consciousness of the openly endless multiplicity of changing experiences and experienced things, one’s own and those of others’.”⁴¹ Any descriptive account of this alleged unity is thus committed to some account of the processes by which temporally consecutive phases of experience form a unity. Thus the phenomenological account of transcendence invokes a structure whose epistemic status is equivalent to posits of naturalistic theories like point particles or valence bonds. Phenomenological transcendence can only be articulated, then, in terms of a process which is ‘transcendent’ in its own terms.

3. Naturalised deconstruction

On the standard reading of Derrida, the trace structure opens consciousness to ‘alterity’ or non-presence. In *Speech and Phenomena* alterity is described as that which is foreign to the intuitive givenness which characterises conscious life.⁴² However, ‘trace’ also names a *radical alterity* or difference which eludes articulation in terms of any set of metaphysical concepts.⁴³ Following Heidegger’s interpretation of the history of philosophy, Derrida frequently characterises the metaphysical *as such* in terms of the notion of ‘presence’.

The ‘metaphysics of presence’ can be understood as a valuation linking the modality of the present and the search for abiding structures or essences which can be articulated in a non-contradictory and complete account of the world.

However, the ‘very idea’ of alterity in Derrida’s work presupposes a constitutive structure to thought or experience which is both transcendental and recalcitrant to conceptualisation. Even if Derrida was *ever* in a position to make such an assumption, my reconstrual of his account of temporality makes it dispensable. Rather than seeking out conditions of possibility for reflectively apparent phenomena, much of Derrida’s writing can be understood to be working through the theoretical consequences of making meaning or content supervene on relations defined within *any* theoretically closed structure – whether these relations involve temporal synthesis (Husserl), intra-linguistic difference (Saussure) or conditions on the applicability or use of words and sentences.

The matrix of concepts defined by trace, differance and iterability describe *in the most general terms* how the contents of signifying states respond to structural and contextual alteration. Derrida abstracts the dynamic elements from Husserl’s account of the temporal constitution of meaning and re-deploys them in an argument schema which (as demonstrated in the synoptic essay ‘Differance’) no longer only pertains to phenomenology but can be applied productively in fields as disparate as semantics and psychoanalysis.⁴⁴ The methodological justification for this is that Husserl is *already and necessarily* engaged in precisely such a speculative theoretical enterprise. The usefulness of this revision (from the naturalistic perspective) is that these schemas can be extended to phenomena where ‘phenomenology’ is not directly at issue. Indeed this is obligatory once we take seriously the implications of the Derridean account of time consciousness. For we have seen that while differance and trace may be ‘phenomenologically derived’ they cannot be phenomenologically interpreted without manifest inconsistency between methodology and ontology. They can be interpreted, if at all, within theoretical frameworks which do not ascribe a foundational role to subjective reflection.

Let us take an example of such an interpretation with generally acknowledged pertinence to materialist science.

Derrida’s reading of Freud’s early work on the neurological basis of memory shows how trace, differance and iterability can be applied and interpreted within naturalistic accounts of cognitive mechanisms. The significance of Freud’s account of memory, for Derrida, lies less in its computational plausibility than in the way it ‘displaces’ a picture of mental representation as a kind of ‘brain writing’ whose inscriptions are unproblematically present or absent. Freud proposed that the resistance to the passage of energy through the neural system exhibited by pathways through a system of ‘psychic’ or ψ neurones

would be *reduced* in proportion to the strength and frequency of stimulation. The 'memory' of past stimulations is represented by the lower resistances associated with frequently stimulated pathways. The scriptural metaphor is warranted in so far as memory is physically 'incised' by the 'breaching' of paths through the ψ system. However, it is impossible to identify this 'trace' with a specific pathway or passage since memory *just is* differences: 'An equality of resistance to breaching, or an equivalence of the breaching forces, would eliminate any preference in the choice of itinerary. Memory would be paralysed. It is the difference between breaches which is the true origin of memory, and thus of psyche'.⁴⁵ In the first instance, then, memory is not represented by absolute quantities in the neural system but by a differential text.

As with the phenomenological trace, these differences are *deferred* rather than belonging to a closed system. The memory trace cannot correspond to a specific ensemble of differences because its *theoretical role* entails the possibility of re-breaching and thus a change in both absolute and relative resistance.⁴⁶ The 'Freudian' trace is thus *formally* equivalent to the 'phenomenological' trace. Yet it is epistemologically distinct because it no longer needs to be treated as an abscess of otherness within the ideal of presence. Indeed, whereas the impossibility of a phenomenological interpretation of trace and difference means that they can only indicate a point of incoherence within the phenomenological project, they can, as Paul Cilliers argues, actively contribute to our understanding of connectionist models of cognition. Just as the Derridean 'trace' is devoid of an essential content, so, Cilliers reminds us:

The significance of a node in a [neural] network is not the result of some characteristic of the node itself; it is the result of the pattern of the weighted inputs and outputs that connect the node to other nodes. The weight, just like the trace, does not stand for anything specific.⁴⁷

As long as we cleave to a representationalist picture according to which abstract contents are assigned to discrete locales (nodes) or states (vectors) within connectionist systems we will fail to do justice to the dynamic character of meaning in 'recurrent' networks which (as in the Lexin model discussed by Van Gelder) are sufficiently complex to allow feedback from current to subsequent states of the system:

The activity of a node is therefore not only determined by its differences from other nodes, but also deferred until its own activity (and those of others) has been reflected back upon it. In this complex pattern of interaction it is impossible to say that a certain sign (or node) represents anything specific.⁴⁸

The deconstructive approach to meaning as simultaneously relational, open to recontextualisation (iteration), and recalcitrant to formalisation is, if Cilliers is right, more pertinent to thinking about content in complex neural networks – and thereby, perhaps, the semiotics of our own biological networks – than one which treats meanings as discrete relations on LOT expressions, or any such ideally repeatable semantic vehicles. Whilst statistical analyses of trained networks uncover repetitions and groupings of states associated with the features the network is trained to detect, these identities are necessary impure and arguably reflect our attempts to manage complexity through formal, functional or intentional interpretation. Thus Jeffery Elman observes that states associated with the ‘same’ words in a recurrent net designed to predict the grammatical role of successive words from sentence order are distributed across sub-regions of the system’s state space according to discriminable variations in semantic or grammatical role (plural or singular; subject or object role; transitive or optional forms of verbs, etc.).⁴⁹ Thus ‘boy’ is represented at a different location depending on whether it functions as direct object (‘boy sees *boy*’) or is the subject of an embedded clause (‘boy who chases *boy* who chases boys’) – where location varies with degree of embedding. Indeed, there could be an indefinite range of tokens ($t_1, t_2 \dots$ etc.) corresponding to fine-grained differences in semantic role (e.g. *burn* in ‘Museum burns’, ‘House burns’).⁵⁰

Such repetitions can only be expected to ‘show up’ within an interpretative stance which applies formal or semantic notions (such as the distinction between word types and tokens) to systems whose complexity allows such refined discriminatory prowess. Iterability can thus be seen as reflecting the way in which types constructed in interpretative metalanguages ‘smear’ across contextual, functional or dynamical differences. Trace and difference, meanwhile, mark the fact that *what* is represented is not decided by the tokening of a type at a given moment or locus but reflects the development of a system in response to the world and to ‘its own activity’. Cilliers’ discussion thus illustrates how ‘holistic’ approaches to content of the kind favoured by Dennett and other interpretationist philosophers of mind entail something like a principle of difference when one allows for the contingency and historicity of the relations which determine the content of significant states (whether these are linguistic, neural, computational, etc.).⁵¹ In consequence, interpretationism – as opposed to intentional realism or, arguably, radical eliminativism – is able to accommodate difference, trace and iterability as meta-theoretical notions since (as Samuel Wheeler has argued with respect to Davidson’s work) it treats any assignment of meaning as a construction within an interpretative idiom, itself subject to alternate interpretations.⁵² The ‘deflationary’ reading of Derrida’s work offered here suggests, in addition, how these notions could

be used to conceptualise relationships between intentional discourse and the subsymbolic processes on which cognition depends.

4. Conclusion

The conclusions of Derridean deconstruction leave no transcendental warrant for the claim that a certain kind of first person perspective is presupposed in any thesis about the world, for the justification for this claim is precisely that objectivity (transcendence) is constituted by the phenomenologically accessible synthesising activity of a temporalised/temporalising transcendental subject. Derrida shows that if this ‘activity’ is anything like the process described in the lectures on time consciousness, it cannot be phenomenologically accessible. The possibility of an eliminativist replacement or reduction of intentional idioms thus cannot be dismissed *a priori* on these grounds. Similarly, if phenomenology is not in a position to satisfactorily unpack notions of objectivity or transcendence, it cannot pretend to explicate the idealised notions of interpretation used in interpretationist accounts.

The ready applicability of deconstructive notions within philosophical accounts of meaning in connectionist networks, as well as in interpretationist accounts generally, also demonstrates that Derrida’s jargon of alterity is inflated as soon as we ‘reinscribe’ his work outside the notional context of transcendental enquiry. In consequence we do not relinquish phenomenological transcendentalism on Derridean grounds only to be committed to a ‘poststructuralist’ or anti-subjectivist version. It is not that the claim that the trace and difference are ‘irreducible to presence or self-presence’ is straightforwardly false. Rather, it is true if ‘presence’ is explicated in phenomenological terms or if understood as shorthand for the presupposition of a classically constituent structure of representation. However, these constraints do not exhaust the metaphysical options available to the contemporary naturalist.

Notes

1. Rodolphe Gasché, *The Tain of the Mirror: Derrida and the Philosophy of Reflection* (London: Harvard University Press, 1986).
2. Richard Rorty articulates this view in opposition to his own ‘neo-pragmatist’ take on Derrida in ‘Is Derrida a Transcendental Philosopher?’, in *Essays on Heidegger and Others*, (Cambridge: CUP, 1991), pp. 119–128.
3. Jacques Derrida *Speech and Phenomena*, trans. David Allison (Evanston Ill.: Northwestern University Press, 1973); *La voix et le phénomène*, (Paris: Presses Universitaires de France, 1967).

4. Some will, no doubt, regard my reading as a 'violent' appropriation of Derrida's texts which ignores their manifestly transcendental (or 'ultra-transcendental') ambitions. This accusation, even if exegetically founded, is simply beside the point. My aim is not fidelity to Derrida's wider philosophical project but to delineate certain aspects of his work which contribute to debates about the place of mind and intentionality in the physical world.
5. The 'otherness', 'alterity' or radical transcendence associated with quasi-transcendental conditions is, I claim, a hangover from the transcendental modes of thinking which deconstruction consistently puts in question (See part 3).
6. See Francisco Calvo Garzón, 'A Connectionist Defence of the Inscrutability Thesis,' *Mind and Language* 15/5 (2000), p 472.
7. Jerry Fodor, *Psychosemantics* (Cambridge Mass.: MIT 1987), p. 97. See also his *A Theory of Content and Other Essays*, (Cambridge Mass.: MIT 1990).
8. Unlike conventional digital computers, neural networks or connectionist systems do not process syntactically structured symbols such as strings of binary code. Their representational content is distributed over weighted patterns of connectivity within populations of neurone-like units. The weights determine how input to the 'hidden units' within the network is transformed into output (how it responds discriminatively to its environment). See, for example, Paul M. Churchland, 'Folk Psychology and the Explanation of Human Behaviour,' *Philosophical Perspectives* 3 (1989), pp. 225–241.
9. Daniel Dennett, 'True Believers: The Intentional Strategy and Why it Works,' in William Lycan, ed. *Mind and Cognition*, (Oxford: Blackwell, 1999), pp. 75–87; 'Beyond Belief,' in *The Intentional Stance*, (Cambridge Mass.: MIT 1987, pp. 117–202.
10. Glossing an untranslated utterance *Gavagai* as 'Lo, a rabbit' (and thus expressive of a belief in a distal rabbit) is appropriate for speakers whose verbal and non-verbal behaviour richly discriminate between animals and non-animals or between different animal kinds; inappropriate otherwise.
11. Donald Davidson, 'Thought and Talk,' in *Inquiries into Truth and Interpretation*, (Oxford: Clarendon Press, 1984), pp. 169–70.
12. Kathleen Akins, 'Of Sensory Systems and the "Aboutness" of Mental States,' *The Journal of Philosophy* 93/7 (1996), pp. 337–372.
13. Terrene Horgan, 'From Supervenience to Superdupervenience: Meeting the Demands of a Material World,' *Mind*, 102/408 (1993) 555–586.
14. J. N. Mohanty, *Transcendental Phenomenology: An Analytic Account*, (Oxford: Blackwell 1989), p. 153.
15. Edmund Husserl, *Ideas: General Introduction to Pure Phenomenology*, trans. W. R. Boyce Gibson (Woking: Unwin, 1931), pp. 147–164.
16. Matthew Ratcliffe, 'Husserl and Nagel on Subjectivity and the Limits of Physical Objectivity,' in *Continental Philosophy Review* 35 2002, p. 364–365. See also Edmund Husserl, *The Crisis of the European Sciences and Transcendental Phenomenology*, trans. David Carr (Evanston: Northwestern University Press, 1970).
17. Paul M. Churchland and Patricia Smith Churchland, 'Stalking the Wild Epistemic Engine,' in *Mind and Cognition*, pp. 212–219; Paul Churchland, 'Conceptual Similarity Across Sensory and Neural Diversity: The Fodor/LePore Challenge Answered,' *Journal of Philosophy* XCV/1 (1998), pp. 5–32.
18. See, for example, 'Husserl and Nagel on Subjectivity and the Limits of Physical Objectivity,' p. 370.

19. *Transcendental Phenomenology*, p. 85.
20. See Juan-José Botero, 'The Immediately Given as Ground and Background,' in Petitot, Varela, Pachoud and Roy, eds. *Naturalising Phenomenology*, (Stanford: Stanford University Press, 1999), pp. 440–463.
21. *Ideas*, section 47.
22. See John Brough, 'Husserl and the Deconstruction of Time,' *Review of Metaphysics* 46 (1993), pp. 503–528; Edmund Husserl, *Phenomenology of Internal Time Consciousness*, trans. James Churchill (The Hague: Martinus Nijhoff, 1982).
23. Maurice Merleau-Ponty, *Phenomenology of Perception*, trans. C. Smith (London: Routledge, 1962), p. 413.
24. *Phenomenology of Internal Time Consciousness*, p. 61.
25. *Phenomenology of Internal Time Consciousness*, pp. 58, p. 72.
26. See David Wood, *The Deconstruction of Time* (Atlantic Highlands NJ.: Humanities Press International, 1989), p. 79.
27. *Speech and Phenomena*, p. 67; *La voix et le phénomène*, p. 75.
28. *Speech and Phenomena* p. 67–68; *La voix et le phénomène*, p. 76; Jacques Derrida, *Of Grammatology*, trans. Gayatri Chakravorty Spivak (London: Johns Hopkins University Press, 1976), p. 184. See also J Claude Evans, *Strategies of Deconstruction: Derrida and the Myth of the Voice* (Minneapolis and Oxford: University of Minnesota Press, 1991), p. 106.
29. *Speech and Phenomena*, p. 88.
30. 'Signature Event Context,' trans. Samuel Weber and Jeffrey Mehlman, in Gerald Graff, ed. *Limited Inc.* (Evanston Ill.: Northwestern University Press, 1988), pp. 1–23.
31. Thus we can think of particular instances of a sign as tokens of that type or particular instances of the belief that Lima is in Peru as being in some relation to the abstract thought or proposition *Lima is in Peru*.
32. Gordon Bearn, 'Differentiating Derrida and Deleuze,' in *Continental Philosophy Review* 33 (2000), 441–465.
33. *Strategies of Deconstruction*, pp. 106–7.
34. Jacques Derrida, *Writing and Difference*, trans. Alan Bass (London: Routledge & Kegan Paul, 1978), p. 163, p. 165.
35. Timothy Mooney, 'Derrida's Empirical Realism,' in *Philosophy and Social Criticism* 25/5 (1999), pp 33–56.
36. *Crisis*, p. 363.
37. *Phenomenology of Internal Time Consciousness*, pp. 62–63.
38. As Paul Ricoeur points out, Husserl is opposed to Kant on this matter. For the latter 'inner sense always falls short of the ability to constitute itself as a source of self-knowledge'. See *Time and Narrative Vol 3*, trans. Kathleen Blamey and David Pellauer (Chicago: University of Chicago Press, 1985), p. 45.
39. See *The Deconstruction of Time*, pp. 78–79.
40. Timothy Van Gelder, 'Wooden Iron, Husserlian Phenomenology Meets Cognitive Science,' in *Naturalising Phenomenology*, p. 260; Robert F. Port, Sven E. Anderson and J. Devin McAuley (1994), 'Towards Audition in an Open Environment'. Retrieved 16 August 2005 from <http://citeseer.ist.psu.edu/339605.html>.
41. *Crisis*, p. 164.
42. *Speech and Phenomena*, p. 65.
43. See, for example, 'Ousia and Gramme,' in *Margins of Philosophy*, trans. Alan Bass (Brighton: Harvester Press, 1982), pp. 63–67.

44. Jacques Derrida, *Margins of Philosophy*, trans. Alan Bass (Brighton: Harvester Press, 1982), pp. 3–27.
45. *Writing and Difference*, p. 201.
46. *Ibid.*, p. 202.
47. Paul Cilliers, *Complexity and Postmodernism: Understanding Complex Systems* (London Routledge, 1998), p. 81.
48. *Ibid.*, p. 82.
49. Jeffery Elman, 'Language as a Dynamical System,' in R Port and T Van Gelder (eds), *Mind as Motion: Explorations in the Dynamics of Cognition*, (MIT: Cambridge Mass., 1995), pp. 208–222.
50. *Ibid.*, pp. 217–219. The binding of temporal sequences in particular states or trajectories here is obviously comparable to the Lexin model.
51. Compare Dennett on the role of functional indeterminacy and the evolution of meanings in *Darwin's Dangerous Idea* (London Penguin, 1995) p. 408.
52. Samuel Wheeler, *Deconstruction as Analytic Philosophy* (Stanford: Stanford University Press, 2000). For a proposed metaphysics of this constructivist approach see David Roden, 'Radical Quotation and Real Repetition,' *Ratio: An international journal of analytic philosophy*, XVII/2 (2004), pp. 191–206.